

Physics 2

Set no. 7

1. In Compton scattering from stationary particles the maximum change in wavelength can be made smaller by using:
 - A. higher frequency radiation
 - B. lower frequency radiation
 - C. more massive express particles
 - D. less massive particles
 - E. particles with greater charge
2. Of the following, Compton scattering from electrons is most easily observed for:
 - A. microwaves
 - B. infrared light
 - C. visible light
 - D. ultraviolet light
 - E. x rays
3. In Compton scattering from stationary electrons the largest change in wavelength occurs when the photon is scattered through:
 - A. 0°
 - B. 22.5°
 - C. 45°
 - D. 90°
 - E. 180°
4. In Compton scattering, the scattered photon has an energy of $E_2 = 120 \text{ keV}$ and the recoiling electron has an energy of $E_3 = 40 \text{ keV}$. Please:
 - a) find the wavelength of the incident photon,
 - b) the angle at which the photon is scattered
5. An incident 71-pm X-ray is incident on a calcite target. Please find the wavelength of the X-ray scattered at a 30° angle.
6. Evidence for the wave nature of matter is:
 - A. electron diffraction experiments of Davisson and Germer
 - B. Thompson's measurement of e/m
 - C. Young's double slit experiment
 - D. the Compton effect
 - E. Lenz's law
7. Which of the following is NOT evidence for the wave nature of matter?
 - A. The photoelectric effect
 - B. The diffraction pattern obtained when electrons pass through a slit
 - C. Electron tunnelling
 - D. The validity of the Heisenberg uncertainty principle
 - E. The interference pattern obtained when electrons pass through a two-slit system

8. Please calculate the longest and the shortest wavelengths in the Balmer series (use Bohr's Model of the Hydrogen Atom)
9. If a hydrogen atom in the ground state absorbs a 93.7-nm photon, corresponding to a transition line in the Lyman series, how does this affect the atom's energy and size (radius)? How much energy is needed to ionize the atom when it is in this excited state?
10. Please calculate the de Broglie wavelength of:
- a 0.65-kg basketball thrown at a speed of 10 m/s,
 - a nonrelativistic electron with a kinetic energy of 1.0 eV.
11. Please find the de Broglie wavelength of an electron in the ground state of hydrogen.
12. If a wave function ψ for a particle moving along the x axis is normalized, then:
- $|\psi|^2 dt = 1$
 - $|\psi|^2 dx = 1$
 - $\partial\psi/\partial x = 1$
 - $\partial\psi/\partial t = 1$
 - $|\psi|^2 = 1$

12. A particle of mass m is constrained to move along a line inside a tube of length L (the tube can be treated as the infinite potential well).

- a) Please justify that the wave function for this particle could be:

$$\psi(x) = \begin{cases} 0 & ; x \in (-\infty, 0) \cup (L, +\infty) \\ A \sin kL & ; x \in (0, L) \end{cases} \quad A, k > 0$$

- Please express k and A constants as a function of L .
- Please find the possible values of energy for this particle.

13. The energy of a particle in a one-dimensional trap with zero potential energy in the interior and infinite potential energy at the walls is proportional to (n = quantum number):

- n
- $1/n$
- $1/n^2$
- $\sqrt{n}$
- n^2

14. The atom is in the excited state for an average of $\Delta t = 10^{-8}$ s. Estimate the uncertainty $\Delta\nu$ of the frequency of emitted photons when the atom transitions from the excited state to the ground state with simultaneous photon emission at an average frequency of $\nu = 7.1 \cdot 10^{14}$ Hz. Is the emitted wave monochromatic?